



200k Chinese accounts shut down

Twitter shut down 200,000 Chinese accounts for spreading disinformation about Hong Kong Protests. <https://digit.in/sep19-55>



Teen finds security flaw

A teen by the name of Bill Demirkapi got suspended for exposing vulnerabilities in his school's software. <https://digit.in/sep19-56>

CNC MACHINES

The usage of CNC machinery tools sprawls across about seven decades, with these machines first being introduced to the workplace after WWII

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omputer Numerical Control or CNC is a revolutionary technology tool where a machine's move-

ments are precisely controlled by computer systems. A CNC Machine comprises of a mini-computer and a microcomputer that effectively controls the machine. The program is stored in the memory of the computer, and the codes within the program can be edited by the controller or programmer as per requirements.

With the earliest forms of CNC Machines being introduced just after World War II, today's CNC machines can operate seamlessly, in a way that capitalises on time and cost. Additionally, it also enables the production of identical, interchangeable parts. Human interaction with these technologically-fueled machines is significantly reduced, which results in far fewer errors. Large fabrication companies follow the practice of leaving such machines running over a long period of time, completely unmanned. The reason behind this is that the software embedded inside these machines stops the machine and calls the operator automatically the instant there is a problem with the system. Let's look at the origin and development of these machines.



BASED ON THE IDEA OF NC OR NUMERICAL CONTROL

While CNC machines are a relatively modern concept in manufacturing and production industries, the concept of these machines harkens back to the basic idea of NC or Numerical Control. The idea of NC stemmed out of the process of automating machine tools which originally incorporated specific concepts of codable logic. However, even before the inception of NC, in 1751, a 'metal-frame turning machine' was made with the primary goal of creating machinery that would

provide greater precision of mechanical operations than what could be achieved using hand-made methods. This essentially represented the beginnings of industrialisation.

Coming back to NCs, the initial machines were fabricated with tools present in the 40s. These tools were then customised using motors that allowed for the movement and control of machinery. The first NC machines were produced in the 1940s, with the more advanced versions starting to show up in the 1950s. In the mid-1940s, John T. Parsons began researching for methods to improve aircrafts by creating stiffened skins for them. A series of Air Force research projects followed, which were conducted at the Massachusetts Institute of Technology (MIT) in 1949. Once the early R&D phases were complete, an experiment milling machine was curated at MIT which involved Professor J.F. Reintjes and his team.

DIFFERENCE BETWEEN NC AND CNC MACHINES

Original NC machines were controlled by following specific points that were embedded on a punched card or



NC system showing machine tool and controller

Credits: Bridgeport Machines Division of Textron Inc.